

Chapter 1 – Introduction to Wazuh

Learning Objectives

After completing this chapter, you will be able to:

- Understand what Wazuh is.
 - Learn why organizations use Wazuh.
 - Understand the history of Wazuh.
 - Identify the main components of the Wazuh platform.
 - Differentiate between SIEM, XDR, and EDR.
 - Recognize the key features of Wazuh.
 - Understand where Wazuh is used in real-world environments.
 - Prepare for the deployment chapters in this guide.
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1.1 What is Wazuh?

Wazuh is a free and open-source security platform used to monitor, detect, and respond to security threats across computers, servers, virtual machines, and cloud environments.

It helps security teams collect logs, detect suspicious activity, monitor system integrity, identify vulnerabilities, and investigate security incidents.

Unlike traditional antivirus software that focuses only on malware, Wazuh provides visibility into the overall security posture of an organization's infrastructure.

Today, Wazuh is widely used by:

- Security Operations Centers (SOC)
- System Administrators
- Blue Team Analysts
- Incident Response Teams

- Government Organizations
 - Educational Institutions
 - Enterprises
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1.2 History of Wazuh

Wazuh was created as a fork of **OSSEC (Open Source Security Event Correlator)**.

OSSEC provided:

- Log Analysis
- File Integrity Monitoring
- Rootkit Detection
- Active Response

As cybersecurity evolved, additional capabilities became necessary.

To meet these requirements, the Wazuh project expanded beyond OSSEC by introducing:

- Vulnerability Detection
- Security Configuration Assessment
- Cloud Security Monitoring
- Malware Detection
- Extended Detection and Response (XDR)
- SIEM Integration
- Modern Web Dashboard

Today, Wazuh is maintained by the **Wazuh Team** and supported by an active open-source community.

1.3 Why Learn Wazuh?

Modern organizations generate millions of security events every day.

Without centralized monitoring, identifying suspicious activity becomes difficult.

Wazuh helps security professionals:

- Collect security logs
- Detect attacks
- Monitor system health
- Identify vulnerabilities
- Investigate incidents
- Generate alerts
- Improve security visibility

For aspiring SOC Analysts, learning Wazuh provides practical experience with enterprise security monitoring.

1.4 Why is Wazuh Important?

Cyberattacks continue to increase in frequency and complexity.

Organizations need tools that can:

- Monitor endpoints
- Detect threats
- Analyze logs
- Alert administrators
- Support investigations

Wazuh combines these capabilities into a single platform.

Instead of checking every server individually, administrators can monitor all connected systems through one centralized dashboard.

Traditional Monitoring vs Wazuh

Without Wazuh

Server A → Server B → Server C → Windows PC → Cloud Serve

Administrator checks every machine separately.

With Wazuh

Server A → Server B → Windows Pc → Cloud Server → Wazuh Manager → Wazuh Dashboard → Security Analyst

1.5 What Does Wazuh Do?

Wazuh continuously monitors systems and collects security-related information.

Examples include:

- Failed login attempts
- Successful logins
- File changes
- Malware detections
- Software vulnerabilities
- User activity
- Service status
- Security policy violations

When suspicious activity is detected, Wazuh generates an alert for the security team.

1.6 Wazuh Features

Wazuh provides numerous security capabilities.

Feature	Description
Log Analysis	Collects and analyzes logs from multiple sources
File Integrity Monitoring (FIM)	Detects unauthorized file modifications
Vulnerability Detection	Identifies vulnerable software
Security Configuration Assessment	Checks system security settings
Rootkit Detection	Detects hidden malicious software
Active Response	Automatically responds to certain threats
Malware Detection	Detects indicators of malware activity
Compliance Monitoring	Helps meet security standards and regulations
Agent Management	Centrally manages monitored endpoints
Dashboard	Provides visualization and reporting

1.7 Understanding SIEM

SIEM stands for:

Security Information and Event Management

A SIEM platform performs two primary functions:

Security Information Management (SIM)

- Collects logs
- Stores logs
- Organizes logs

Security Event Management (SEM)

- Detects suspicious activity
- Correlates events
- Generates alerts

- Supports investigations

Together, these capabilities help analysts identify security incidents more efficiently.

1.8 SIEM vs EDR vs XDR

Many beginners confuse these technologies.

Although they work together, each has a different purpose.

Technology	Purpose
SIEM	Collects, stores, and analyzes security logs
EDR	Detects and responds to threats on endpoints
XDR	Correlates security data across multiple security products

SIEM

Focuses on:

- Logs
- Alerts
- Correlation
- Monitoring

Example:

Firewall Logs→Windows Logs→Linux Logs→SIEM

EDR

Focuses on endpoint protection.

Example:

Laptop→Detect Malware→Kill Process→Quarantine File

XDR

Combines multiple security technologies.

Example:

Firewall→Email→Endpoint→Cloud→Identity→Single Detection Platform

1.9 Wazuh Components

A Wazuh deployment consists of several components working together.

Component	Purpose
Wazuh Manager	Processes security events and manages agents
Wazuh Indexer	Stores indexed security data
Wazuh Dashboard	Web interface for monitoring and investigation
Wazuh Agent	Collects logs and system information from endpoints

We will study each component in detail in Chapter 2.

1.10 Supported Operating Systems

Wazuh supports a wide range of operating systems.

Server

- Ubuntu Server
- Debian
- Red Hat Enterprise Linux
- Rocky Linux
- AlmaLinux

Agents

- Windows
- Linux
- macOS

Cloud

- AWS
 - Microsoft Azure
 - Google Cloud Platform
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1.11 Real-World Use Cases

Organizations use Wazuh for:

- Security Monitoring
 - Threat Detection
 - Compliance Monitoring
 - Incident Response
 - File Integrity Monitoring
 - Vulnerability Management
 - Malware Detection
 - Log Analysis
 - Cloud Security
 - Endpoint Monitoring
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1.12 Wazuh in a Blue Team Home Lab

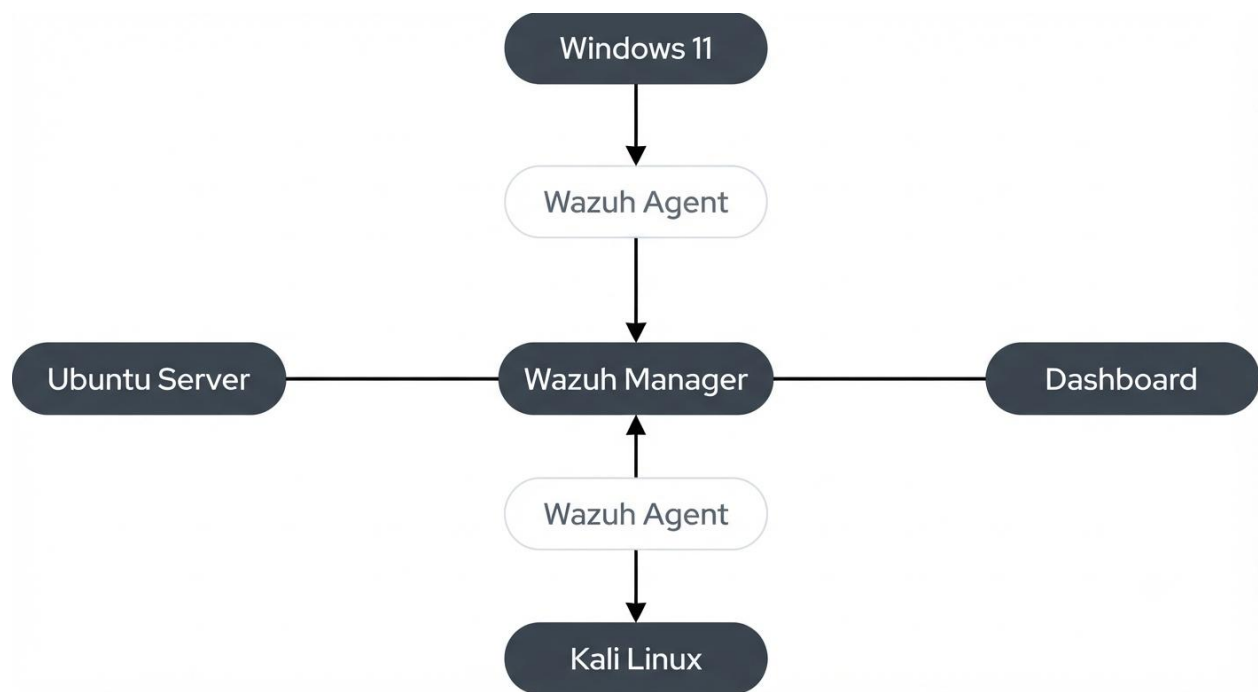
In this handbook, Wazuh will serve as the central monitoring platform.

Our lab will consist of:

- Ubuntu Server (Wazuh Server)
- Windows 11
- Kali Linux

Future guides will connect these systems to Wazuh for monitoring and analysis.

Home Lab Overview



1.13 Ethical and Legal Considerations

Wazuh is designed to improve security visibility and incident detection.

Always use Wazuh:

- On systems you own.
- In authorized lab environments.
- With proper organizational approval.
- According to your organization's security policies.

Do not monitor systems without authorization.

Best Practices

- Understand the purpose of each Wazuh component before deployment.
 - Learn the difference between SIEM, EDR, and XDR.
 - Build and test Wazuh in a home lab before deploying it in production.
 - Follow the official Wazuh documentation for version-specific changes.
 - Keep Ubuntu Server updated before installing Wazuh.
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Chapter Summary

In this chapter, you learned:

- What Wazuh is and why it is widely used.
- The history and evolution of the Wazuh project.
- The purpose of SIEM, EDR, and XDR.
- The core components of a Wazuh deployment.
- The main security capabilities provided by Wazuh.
- Common real-world use cases.
- How Wazuh fits into a Blue Team home lab.

This foundation will help you understand the deployment and configuration steps covered in the following chapters.

One-Line Summary

Wazuh is an open-source security platform that combines SIEM and XDR capabilities to collect, analyze, and respond to security events across servers, workstations, and cloud environments, making it a powerful solution for modern Blue Team operations.
